

SEDA: Spectral Energy Distribution Analyzer for Forward Modeling and Empirical Analysis of Ultracool Objects

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Software

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Summary

Brown dwarfs and extrasolar gas giant planets exhibit complex, dynamic atmospheres that undergo diverse physical and chemical processes, including disequilibrium chemistry, cloud formation, auroral activity, and variability, which depend on fundamental properties and observational factors such as temperature, age, metallicity, rotation period, and viewing geometry (Biller, 2017; e.g., M. S. Marley & Robinson, 2015; Vos et al., 2017). These atmospheres also exhibit a rich diversity of chemical species, including water, methane,

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45 ammonia, carbon monoxide, carbon dioxide, and silicates (Cushing et al., 2006; e.g., Kirkpatrick,
46 2005). Studying the chemistry and physics of these atmospheres provides insights not only
47 into the atmospheric processes that govern their behavior, but also into the formation and
48 evolution of these objects (e.g., Luhman, 2012). In the long term, understanding these
49 atmospheric conditions is essential for characterizing potential habitable worlds with next-
50 generation observatories and for contextualizing the uniqueness of our own.

51 We introduce SEDA (Spectral Energy Distribution Analyzer), a versatile open-source Python
52 package for forward modeling of observations of ultracool objects, such as brown dwarfs,
53 directly imaged exoplanets, and low-mass stars. This workflow employs a Bayesian framework
54 based on pre-computed, self-consistent atmospheric models. SEDA also incorporates a broad
55 set of empirical techniques and auxiliary tools for spectral energy distribution (SED) analysis,
56 as described in Functionality. Overall, SEDA enables the inference and analysis of atmospheric
57 composition, cloud properties, and fundamental physical parameters of ultracool objects from
58 observational data.

59 Statement of Need

60 The James Webb Space Telescope (JWST) is revolutionizing our understanding of extrasolar
61 atmospheres by providing high-quality observations of exoplanets and brown dwarfs. These data
62 will synergize with observations from next-generation facilities such as the Nancy Grace Roman
63 Space Telescope, the Habitable Worlds Observatory, the Vera C. Rubin Observatory, and 30-m
64 class telescopes (e.g., the Extremely Large Telescope, ELT, and the Thirty Meter Telescope,
65 TMT). Furthermore, a large volume of multiwavelength data is already available from wide-field
66 photometric surveys (e.g., SDSS, York et al., 2000; and Pan-STARRS, Chambers et al., 2016),
67 all-sky photometric surveys (e.g., Gaia, Gaia Collaboration et al., 2020; 2MASS, Skrutskie
68 et al., 2006; and WISE, Cutri et al., 2013), targeted spectroscopic programs (e.g., SpeX
69 Prism Library, Adam J. Burgasser, 2014; MOCAdB, Gagné et al., 2026), and both large (e.g.,
70 LAMOST, Luo et al., 2015; and APOGEE, Abdurro'uf et al., 2022) and all-sky (SPHEREx,
71 Doré et al., 2014) spectroscopic surveys.

72 To fully exploit these extensive datasets, it is necessary to develop techniques and software
73 capable of managing heterogeneous, multiwavelength observations (see State of the Field).
74 SEDA contributes to this need by providing a framework for the analysis and forward modeling of
75 photometric and spectroscopic multiwavelength data either separately or jointly, using spectra
76 of varying resolution and/or overlapping wavelength coverage, as well as photometry from a
77 broad range of filter systems compiled by the Spanish Virtual Observatory (SVO) Filter Profile
78 Service¹. In addition to its modeling capabilities, the package incorporates a growing set of
79 empirical techniques commonly used in the literature, offering a unified and flexible framework
80 for both model-based and empirical SED analysis (see Functionality).

81 State of the Field

82 Several software packages enable forward modeling of ultracool dwarf data, including VOSA
83 (Bayo et al., 2008), Starfish (Czekala et al., 2015), SEDkit (Filippazzo et al., 2015), SPLAT
84 (A. J. Burgasser & Splat Development Team, 2017), species (Stolker et al., 2020), ForMoSA
85 (Petrus et al., 2023), PICASO (JWST Transiting Exoplanet Community Early Release Science
86 Team et al., 2023), and ucdmcmc (A. Burgasser et al., 2026). Except for VOSA, these packages
87 primarily focus on spectroscopic data, with limited ability to incorporate photometry in the
88 model fitting. To our knowledge, only species can forward model SEDs constructed from
89 photometry alone or combined with spectra, but its empirical analysis capabilities are relatively
90 limited. In contrast, SPLAT provides more empirical analysis tools similar to those in SEDA, but
91 it is restricted to modeling spectroscopic observations. SEDA complements these packages by

¹<https://svo2.cab.inta-csic.es/theory/fps/>

supporting the modeling of spectrophotometric SEDs and offering extensive empirical analysis techniques, facilitating the study of brown dwarfs and directly imaged exoplanets.

Functionality

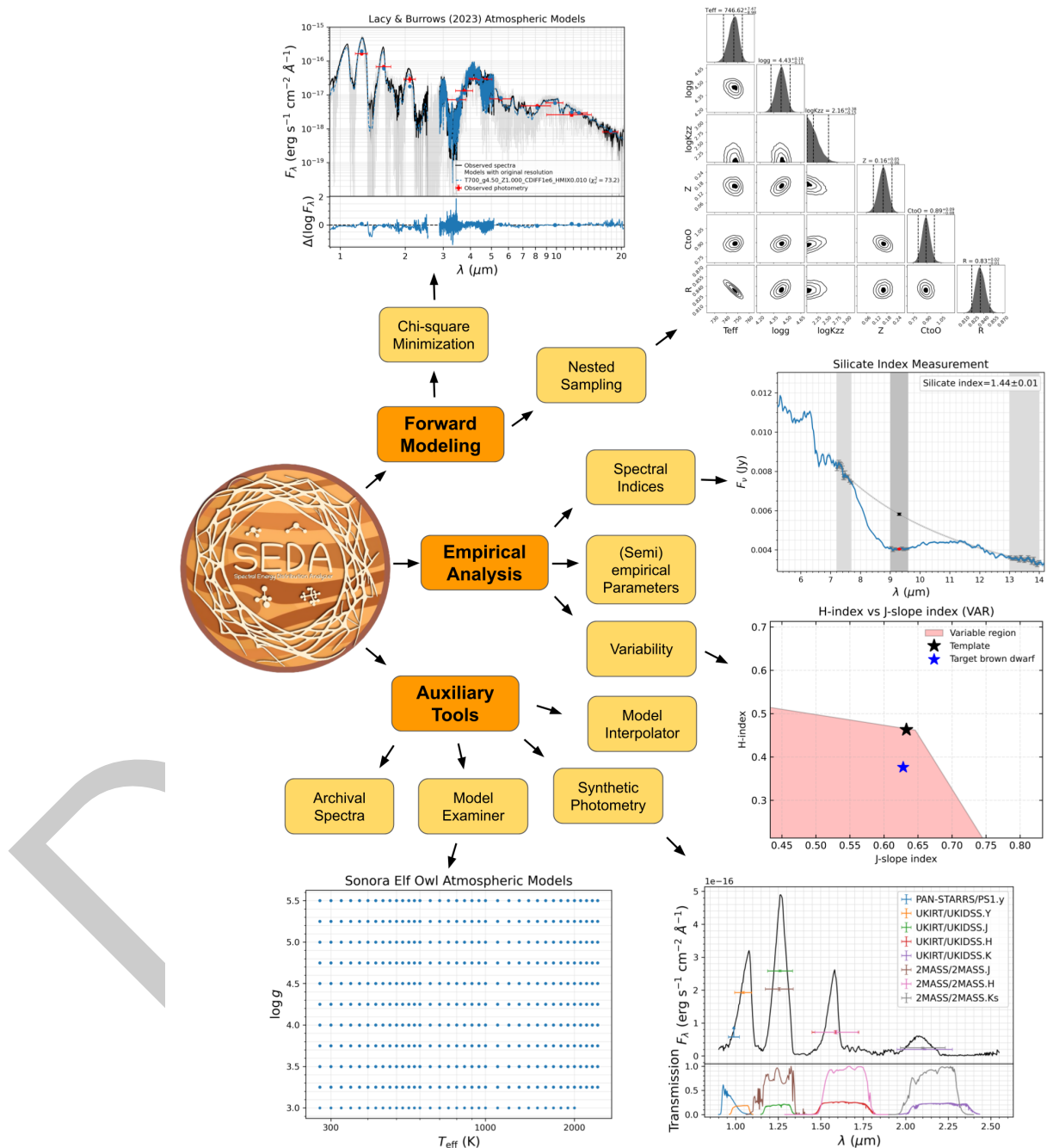


Figure 1: Overview of the SEDA workflow, illustrating key components such as forward modeling, empirical analysis, and additional useful tools.

95 [Figure 1](#) illustrates the core functionality and key tools of SEDA.

96 SEDA forward modeling employs two complementary approaches: a Bayesian framework with
 97 nested sampling to construct posterior distributions and a chi-square minimization method
 98 to find the best-fitting models (see [Mathematics](#)). The code supports both widely used and
 99 state-of-the-art atmospheric model grids, namely Lacy (2026, in preparation), SAND ([Alvarado
 100 et al., 2024](#)), Sonora Diamondback ([Morley et al., 2024](#)), Sonora Elf Owl ([Mukherjee et al.,
 101 2024](#)), Lacy & Burrows (2023), Sonora Cholla ([Karalidi et al., 2021](#)), Sonora Bobcat ([Mark S.
 102 Marley et al., 2021](#)), ATMO 2020 ([Phillips et al., 2020](#)), BT-Settl ([Allard et al., 2012](#)), and
 103 Saumon & Marley (2008). Collectively, these models span a broad range of atmospheric and
 104 physical conditions, from cloud-free to cloudy atmospheres; equilibrium and non-equilibrium
 105 chemistry; metallicities from very low (-2.4 dex) to super-solar ($+0.5$ dex); temperatures from
 106 the coldest brown dwarfs through hot and temperate exoplanets to low-mass stars ($\sim 300\text{--}4200$
 107 K); and surface gravities ($\log g=3.0\text{--}5.5$) appropriate for both young and old ultracool objects.
 108 SEDA also allows users to easily incorporate additional atmospheric model grids, as well as
 109 evolutionary models. Fitting these atmospheric model grids to observational data, the code
 110 infers chemical and physical parameters defined as free parameters in the grids, along with
 111 additional quantities such as bolometric luminosity and radius, when the distance is known
 112 (see [Mathematics](#)). Additional fundamental parameters, including mass, age, surface gravity,
 113 and effective temperature, can be derived using evolutionary models from the model families
 114 described above, as well as BHAC15 ([Baraffe et al., 2015](#)).

115 Empirical analyses of SEDs include measurements of the strengths of key spectral features
 116 produced by gas and grain species using literature-defined or user-defined spectral indices, i.e.,
 117 flux ratios that quantify the depth of specific spectral signatures. These indices are applied to
 118 mid-infrared features (e.g., [Cushing et al., 2006](#); [Suárez & Metchev, 2022, 2023](#)), as well as
 119 to near-infrared diagnostics designed to identify potential variable objects (e.g., [Ashraf et al.,
 120 2022](#); [Oliveros-Gomez et al., 2022, 2024](#)). Additional empirical analyses include the estimation
 121 of fundamental parameters such as bolometric luminosity and radius, and the application of
 122 published relationships between fundamental parameters, including temperature-spectral type,
 123 luminosity-spectral type, and inclination-cloudiness-color trends (e.g., [Filippazzo et al., 2015](#);
 124 [Kirkpatrick et al., 2021](#); [Suárez et al., 2023](#); [Vos et al., 2017](#)).

125 In addition, SEDA provides several auxiliary tools that can be used independently, including: (i)
 126 the calculation of synthetic photometry from spectra for any filter available through SVO, (ii) a
 127 model interpolator that generates synthetic spectra for any parameter combinations within the
 128 grid coverage of available atmospheric models, (iii) a model examiner that enables inspection
 129 of atmospheric model properties, such as basic and free parameters, parameter coverage,
 130 resolving power, and spectral resolution, and (iv) access to 113 Spitzer IRS mid-infrared
 131 spectra of brown dwarfs from [Suárez & Metchev \(2022\)](#), together with associated target
 132 properties, observing logs, and measured spectral indices.

133 Mathematics

134 For model fitting, as described in [Functionality](#), the code adopts two approaches:

135 1. Chi-square Minimization

136 The code uses the following weighted reduced chi-square (χ_r^2) definition:

$$\chi_r^2 = \frac{1}{N - n_p} \sum_{i=1}^N w_i \left(\frac{F_{\text{obs},i} - \alpha F_{\text{mod},i}}{\sigma_i} \right)^2 \quad (1)$$

137 Here, N is the number of fitted data points, n_p is the number of free model parameters (so
 138 $N - n_p$ are the degrees of freedom), w_i is the weight assigned to each data point to balance
 139 the relative contributions of datasets with different numbers of data points, $F_{\text{obs},i}$ and σ_i are

the observed fluxes and their uncertainties, $F_{\text{mod},i}$ are the model fluxes, and α is an unknown geometric dilution factor equal to $(R/d)^2$, where R is the object radius and d is its distance.

The code provides several options for assigning the weights w_i , which are designed to compensate for differences in the number of data points among datasets and to ensure no dataset has a negligible contribution to the fit (e.g., photometric data versus spectra, or low-resolution versus high-resolution spectra). These options include: (i) assigning each dataset a weight equal to the inverse of its total number of data points, so that all datasets contribute the same, even if they have very different numbers of points; and (ii) weighting spectroscopic data points by their wavelength resolution (i.e., wavelength step) and photometric data points by the filter effective width, such that broader filters or low-resolution spectra receive larger weights than high-resolution spectra.

To determine the value of α that minimizes χ_r^2 , the code can either use the non-linear least-squares minimization python package LMFIT (Newville et al., 2014) or compute the analytic solution obtained by setting $\frac{\partial \chi_r^2}{\partial \alpha} = 0$, which is given by:

$$\alpha = \frac{\sum_{i=1}^N w_i \frac{F_{\text{obs},i} F_{\text{mod},i}}{\sigma_i^2}}{\sum_{i=1}^N w_i \frac{F_{\text{mod},i}^2}{\sigma_i^2}} \quad (2)$$

2. Nested Sampling

For Bayesian model fitting, the code adopts the weighted Gaussian log-likelihood ($\ln \mathcal{L}$):

$$\ln \mathcal{L} = -\frac{1}{2} \sum_{i=1}^N \left[w_i \frac{(F_{\text{obs},i} - \alpha F_{\text{mod},i})^2}{\sigma_i^2} + \ln(2\pi \sigma_i^2) \right] \quad (3)$$

with all parameters defined as in Equation 1. The code wraps the model interpolator with the dynamic nested sampling package DYNESTY (Speagle, 2020) to construct posteriors. During sampling, each generated synthetic spectrum is scaled using either $(R/d)^2$ if distance is known, or the factor that maximizes $\ln \mathcal{L}$, which is the same that minimizes χ_r^2 (Equation 2).

The scaling factor α is used to derive the radius from the equation above, independently of evolutionary models, provided that the distance is known. The observed data are complemented with the best-fitting model to construct a full hybrid SED, which is then integrated to determine the bolometric luminosity, if the distance is available. The radius and bolometric luminosity are used to estimate the effective temperature via the Stefan-Boltzmann law and can also be interpolated within evolutionary models to derive additional parameters, including mass, surface gravity, and age.

Documentation

SEDA is hosted on GitHub² and the most up-to-date documentation is available on Read The Docs³. The documentation provides installation instructions, an overview of the code and its principal modules, a description of the code workflow, a list of useful tools, and details on the available atmospheric models. The repository also provides Jupyter notebook tutorials demonstrating typical scientific workflows.

The documentation is actively maintained and includes an API reference describing all functions and classes, with basic example provided where applicable. In addition, the documentation

²<https://github.com/suarezgenaro/seda>

³<https://seda.readthedocs.io/>

175 contains a frequently asked questions section and guidance on contributing to the code or
176 submitting feedback.

177 Software Design

178 SEDA v1.0 accompanies this publication and represents a stable, well-tested release following
179 multiple pre-releases used for development and community feedback. The software is designed
180 for flexible analysis of SEDs, centered on forward modeling and complemented by empirical
181 analysis tools (see [Functionality](#) and [Mathematics](#)). The code includes automated tests to
182 verify core functionality and has been successfully tested with Python 3.9 or later (latest tested:
183 Python 3.14) on Linux, Windows, and macOS.

184 Research Impact

185 The foundation of SEDA was introduced in Suárez et al. (2021), which implemented only a
186 chi-square minimization approach. In this work, we substantially expand upon that initial
187 framework and release a significantly enhanced version of the software to the community.
188 SEDA has already been used in at least 14 publications (including published, submitted, or in
189 preparation works) (Kiman et al., 2026; L'Heureux et al., 2026; Lam et al., 2026; Rothermich
190 et al., 2026; e.g., Suárez et al., 2025).

191 Future Developments

192 Among several planned additions that will further increase the functionality of the code, future
193 releases will (i) enable measurements of projected rotational velocity ($v \sin i$) and radial velocity
194 (RV) by incorporating rotational broadening and Doppler shifting into atmospheric model
195 spectra, as well as inference of viewing geometry based on $v \sin i$, rotation period, and radius
196 constraints (e.g., Vos et al., 2017), and (ii) implement cross-correlation analyses to identify
197 molecular species in medium- to high-resolution spectroscopic observations (e.g., Petit dit de
198 la Roche et al., 2018).

199 We welcome feedback, suggestions, and contributions from the community to help improve
200 SEDA in future releases.

201 AI Usage Disclosure

202 Generative AI tools (ChatGPT and Gemini) were used in a limited capacity to help polish the
203 language of the documentation and to provide occasional coding suggestions or simplifications.
204 All software design, implementation, and scientific decisions were developed by the authors.
205 Any AI-generated suggestions were reviewed, verified, and modified by the authors before
206 inclusion.

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